SMART HEALTHCARE DATA ANALYSIS DASHBOARD

Tableau-Based Data Visualization Project Report

Project Title	Smart Healthcare Data Analysis Dashboard
Visualization Tool	Tableau
Dataset	smart_healthcare_dataset.csv
Dataset Size	5,000 records × 16 columns
Workbook Files	teblue proj.twb and teblue proj1.twbx

Prepared as project documentation for GitHub publication

1. Abstract

The Smart Healthcare Data Analysis Dashboard is a Tableau-based data visualization project. It explores a healthcare dataset containing 5,000 patient records and 16 attributes. The workbook presents views of patient demographics, body mass index (BMI), blood pressure, glucose, cholesterol, lifestyle indicators, symptoms, and heart-disease labels. The goal is to organize the data into visual summaries that make patterns easier to explore. The project is descriptive and educational; it is not intended to diagnose disease or recommend treatment.

2. Introduction

Healthcare datasets often contain a mixture of demographic information, clinical measurements, lifestyle variables, and health-condition labels. Reviewing these variables in a tabular format can make comparisons difficult. Data visualization helps users explore distributions and compare groups. This project uses Tableau worksheets, a dashboard, and a story to present selected views of the supplied healthcare dataset.

3. Problem Statement

The project addresses the difficulty of exploring multiple healthcare attributes in a single, understandable view. It aims to provide visual summaries of patient demographics and selected health and lifestyle indicators, and to make it easier to compare these attributes with the heart_disease field.

4. Objectives

• Analyze patient distribution by age and gender.

• Explore BMI, blood pressure, glucose, and cholesterol in relation to the heart-disease label.

• Visualize lifestyle indicators such as exercise level and smoking.

• Review symptom indicators including fatigue, chest pain, and dizziness.

• Build a Tableau dashboard/story that brings the visualizations together.

5. Tools and Technologies

• Tableau: creating worksheets, charts, dashboard layouts, and a story.

• CSV: storing the tabular healthcare dataset.

• GitHub: publishing and version-controlling the project files.

6. Dataset Description

The packaged Tableau workbook contains a CSV file named smart_healthcare_dataset.csv. It has 5,000 rows and 16 columns. The attributes cover age, gender, BMI, exercise level, smoking, alcohol use, blood pressure, cholesterol, glucose, fatigue, chest pain, dizziness, heart disease, diabetes, stroke, and health_risk_score.

Attribute	Meaning in supplied file
age	Age in years
gender	Gender category
bmi	Body Mass Index value
exercise_level	Encoded exercise-level value
smoking / alcohol	Lifestyle indicator fields
blood_pressure / cholesterol / glucose	Numeric health measurements
fatigue / chest_pain / dizziness	Symptom indicator fields

Attribute	Meaning in supplied file
heart_disease / diabetes / stroke	Condition label fields
health_risk_score	Health-risk score field

The workbook does not provide a complete data dictionary for every encoded value. Therefore, the exact meaning of each numeric code should be confirmed from the dataset's original documentation before interpreting it.

7. Methodology

• Opened and inspected the Tableau workbook and its packaged data source.

• Identified the available fields and the nine worksheets in the workbook.

• Reviewed the CSV structure and calculated basic descriptive counts for documentation.

• Organized the analysis into demographic, clinical-measurement, lifestyle, symptom, and disease-label views.

• Presented the findings as descriptive exploration rather than causal or clinical conclusions.

8. Dashboard and Worksheet Description

Worksheet	Purpose
BMI vs Disease	Explores BMI category alongside the heart-disease field.
BP vs Glucose	Compares blood pressure and glucose, grouped by heart-disease label.
Cholesterol by Disease	Shows cholesterol distribution by heart-disease category.
Exercise vs Disease Risk	Explores exercise level against the heart-disease measure.
Heart Disease by Age	Examines heart-disease records across age groups.
Lifestyle Risk Matrix	Combines lifestyle-related dimensions with the heart-disease measure.
Smoking Distribution Circle	Shows smoking categories using record counts.
Symptom Analysis	Brings multiple symptom measures together for comparison.
Patients by Gender	Summarizes patient records by gender.

The workbook also contains a dashboard named Dashboard 1 and a story named Story 1. The worksheet names above are taken from the supplied Tableau workbook.

9. Descriptive Dataset Snapshot

Measure	Observed value
Number of records	5,000
Number of columns	16
Age range	18–79 years
Average age	48.8 years
Female records	2,506
Male records	2,494
heart_disease = 1	1,499 (30.0%)
heart_disease = 0	3,501 (70.0%)

These figures describe only the supplied dataset. They are not estimates for a general or clinical population.

10. Data Quality Notes and Limitations

• **Constant score field:** health_risk_score equals 100 in all 5,000 records, so it has zero variation and cannot distinguish risk levels. This field should be checked against the original data source.

• **Encoded fields:** the workbook does not explain all 0/1 or exercise-level codes. A verified data dictionary is needed for reliable interpretation.

• **Unknown provenance:** the workbook does not document the dataset's collection method, population, or source.

• **Association is not causation:** visual comparisons do not establish that a lifestyle or clinical factor causes a condition.

• **Educational use:** the dashboard is not a medical diagnostic, screening, or treatment system.

11. How to Run the Project

• Download or clone the GitHub repository.

• Open teblue proj1.twbx in Tableau Desktop or a compatible Tableau application. The packaged workbook is the recommended file to start with.

• If opening teblue proj.twb, reconnect the CSV data source if Tableau reports a missing file.

• Open Dashboard 1 to view the dashboard, or select individual worksheets for detailed exploration.

• Use the Story 1 tab if you want to navigate the workbook's story presentation.

12. Conclusion

The Smart Healthcare Data Analysis Dashboard demonstrates how Tableau can turn a multi-column healthcare dataset into a set of visual summaries. Its nine worksheets cover demographics, clinical measurements, lifestyle factors, symptoms, and disease labels, with a dashboard and story for presentation. The project provides a useful foundation for exploratory analysis. Before extending the analysis, the dataset's provenance, encoded values, and constant health_risk_score field should be validated.

13. GitHub Repository Contents

• teblue proj.twb – Tableau workbook definition.

• teblue proj1.twbx – packaged Tableau workbook containing the CSV data source.

• Project_Report.pdf – this project report.

• README.md – repository overview and opening instructions.

Suggested repository title: Smart-Healthcare-Data-Analysis-Dashboard

Suggested GitHub description: Tableau dashboard exploring patient demographics, clinical indicators, lifestyle factors, symptoms, and heart-disease labels.